

FATHIMA SHIHANA SALAHUDEEN

Research Scientist | Translational Toxicology | RNA Biomarkers

📞 0421 368420 ✉ fathimashihana20@gmail.com [linkedin.com](#) 📍 24 Benaroon Road, Lakemba NSW 2195

SUMMARY

A highly accomplished Research Scientist with over a decade of experience in toxicology, molecular biology, and biomedical research. Expert in zebrafish disease models, CRISPR-Cas9 gene editing, and microRNA biomarker discovery, with a strong track record of high-impact publications and international presentations. Dedicated to advancing human health through innovative translational research, interdisciplinary collaboration, and data-driven scientific inquiry.

WORK EXPERIENCE

Researcher - Voluntary
Centenary Institute of Cancer Medicine & Cell Biology

📅 01/2026 - Present 📍 Sydney, Australia

- Analyse data for ongoing research projects
- Support scientific manuscript writing and editing
- Assist in grant application preparation
- Collaborate with multidisciplinary research teams

Post-Doctoral Researcher
Centenary Institute of Cancer Medicine & Cell Biology

📅 11/2024 - 12/2025 📍 Sydney, Australia

- Led preclinical liver disease research using zebrafish models, performing model induction, phenotypic assessments, and lab-based experiments; contributed to experimental design, literature reviews, and statistical analysis to drive data-driven results.
- Collaborated across disciplines with molecular biologists, clinicians, and data scientists to deliver translational research outcomes; presented findings at national and international conferences and published peer-reviewed papers.

Research Officer
Centenary Institute of Cancer Medicine & Cell Biology

📅 03/2021 - 12/2023 📍 Sydney, Australia

- Designed and conducted experimental studies using zebrafish models to investigate alcoholic and non-alcoholic fatty liver disease, contributing to advancements in hepatology research and producing two peer-reviewed publications.
- Led interdisciplinary collaborations to develop and execute research protocols, integrating molecular biology, imaging, and toxicology approaches to strengthen project outcomes.
- Incorporated statistical analysis and literature synthesis to support hypothesis-driven research, enabling successful grant applications and attracting new research funding.
- Supervised and mentored undergraduate summer research students, enhancing lab productivity and fostering skill development in emerging scientists.
- Presented findings at national conferences and in high-impact journals, contributing to the global discourse on liver disease and microRNA biomarker research.

KEY ACHIEVEMENTS

- 📄 **Published in Leading Journals**
Authored 28+ peer-reviewed papers, including first-author studies on liver disease, toxicology, and microRNA biomarkers.
- ★ **Zebrafish Model Innovator**
Developed zebrafish models for fatty liver disease, providing key insights into lipid gene functions.
- 🧬 **CRISPR Gene Editing Expert**
Used CRISPR-Cas9 to create genetic knockdowns, identifying genes linked to alcohol-induced liver injury.
- 🩺 **Low-Cost Diagnostic Developer**
Created a bedside test for methemoglobinemia, now used internationally to guide patient care.
- 👤 **Research Mentorship Leader**
Supervised MPhil, Honours, and summer students —many now pursuing research careers.

SKILLS

- Zebrafish disease modelling (alcoholic & non-alcoholic fatty liver)
- CRISPR-Cas9 gene editing
- MicroRNA biomarker analysis
- Fluorescence and confocal microscopy
- Flow cytometry
- ELISA and spectrophotometry
- RT-qPCR and high-throughput miRNA analysis
- Human sample processing and pharmacokinetics
- Toxicology and nephrotoxicity research
- Clinical data collection and analysis
- Scientific writing and peer-reviewed publication
- Grant writing and funding acquisition
- Laboratory and project management
- Student supervision and mentorship
- Adherence to animal ethics and biosafety protocols
- Conference presentations and public speaking
- Multidisciplinary research collaboration

AFFILIATIONS & CERTIFICATES

- Research Affiliate—University of Sydney (from 2020 to Present)
- Applied Project Management (2025)
- ICH - GCP Training Certificate (2024)
- Data Science with R (2021)
- Data Science with Python (2021)

Research Assistant
Centenary Institute of Cancer Medicine & Cell Biology

📅 02/2020 - 03/2021 📍 Sydney, Australia

- Developed CRISPR-Cas9 zebrafish models to investigate gene function in alcohol-induced liver injury, enabling the identification of key genetic contributors to hepatic damage.
- Managed end-to-end project workflows, including experimental setup, data collection, statistical analysis, and reporting, ensuring rigorous and reproducible results.
- Oversaw participant recruitment and ethics compliance, upholding institutional standards and research integrity across all study phases.

Research Associate
South Asian Clinical Toxicology Research Collaboration

📅 06/2013 - 11/2016 📍 Peradeniya, Sri Lanka

- Led laboratory operations in the Renal Biomarker Program as team leader.
- Supervised three honours students, mentored two MPhil candidates, and trained seven laboratory analysts.
- Handled and processed clinical samples while maintaining strict adherence to regulatory and ethical compliance.
- Ensured accurate documentation and reporting of test results, and provided detailed analyses to Principal Investigators.
- Oversaw general laboratory housekeeping, including cleaning, maintenance, and calibration of lab equipment.
- Maintained laboratory and storage facilities according to national safety and quality standards.
- Managed inventory, ordering, and tracking of lab supplies to ensure uninterrupted workflow.

Research Assistant
South Asian Clinical Toxicology Research Collaboration

📅 09/2010 - 05/2013 📍 Peradeniya, Sri Lanka

- Analysed samples, interpreted data and reported scientific results adhering to strict regulations
- Worked collaboratively in cross-functional teams
- Standardised and performed all ELISA tests and designed SOP for all tests
- Standardised and undertaking of AChE developed bedside tests and assay

EDUCATION

Doctor of Philosophy - Medicine
The University of Sydney

📅 02/2017 - 03/2021

Title: MicroRNAs Biomarkers in Human Nephrotoxicity

Master of Philosophy - Medicine
University of Peradeniya

📅 09/2009 - 02/2011

Title: Development and Validation of Accessible Laboratory Tests to Inform the Clinical Management Self-Poisonings

Bachelor of Business - Chemistry
University of Peradeniya

📅 06/2002 - 06/2006

GPA - 3.49

SCHOLARSHIPS AND AWARDS

- APAMT Travel Scholarship – Malaysia (2019)
- Best Oral Presentation Award – APAMT Conference, Malaysia (2019)
- Presidential Award for Scientific Publication – Government of Sri Lanka (2019, 2018, 2017, 2016, 2013, 2010)
- NHMRC Translational Clinical Toxicology Travel Scholarship – Australia (2018, multiple rounds)
- University of Sydney International Research Scholarship – Australia (2017–2020)
- Postgraduate Research Scholarship – The Hospital Research Foundation (2017)
- SACTRC Travel Scholarship – 14th APAMT Congress, Australia (2015)
- National Research Council Merit Award for Scientific Publication – Sri Lanka
- MPhil Research Funding – South Asian Clinical Toxicology Research Collaboration (2007–2010)
- APAMT Travel Scholarships – Vietnam (2010), India (2008), Thailand (2007)
- International Travel Award – NACCT Conference, Canada (2008)

REFEREES

Prof. Nicholas A. Buckley

Professor of Clinical Pharmacology

The University of Sydney, NSW, Australia

☎ (+61) 2 9351 3819

✉ nicholas.buckley@sydney.edu.au

Prof. Devanshi Seth

Principal Scientist

The University of Sydney, NSW, Australia

☎ (+61) 2 9565 6268

✉ d.seth@sydney.edu.au

Prof. Andrew H. Dawson

Clinical Professor

Royal Prince Alfred Hospital, NSW, Australia

☎ (+61) 2 9515 6419

✉ ahdawson@gmail.com

REVIEWED PEER-REVIEWED JOURNALS

- Nature Scientific Reports – 2022 to Present (4 reviewed papers)
- A General Method for the Construction of Plasma Standard Curves as used in Colorimetric Assays, MethodsX, 2018 SingleDose Pharmacokinetics of Metoclopramide Oral Tablets Utilizing HPLC-UV Method, Current Pharmaceutical Analysis, 2018
- Stability of post-mortem methemoglobin: Artifactual changes caused by storage conditions, Forensic Science International. 2018 Feb; 283: 21-28. doi: 10.1016/j.forsciint.2017
- Application of Validated Stability-Indicating HPLC Method in Stability Testing of Acetaminophen and Guaiphenesin Tablets, Chem Sci Trans., 2014, 3(1), pp 201-208
- Stability of methemoglobin in blood samples from normal individuals and patients treated with dapsone, Clinical Chemistry and Laboratory Medicine, 2013

PEER PUBLICATIONS

- 2026 - Nicole Coelho, Lopes, Ranieri Sales, **Shihana F**, et al. Role of Serum IL-33 in Bothrops Snakebite Victims: Linking Inflammation and Endothelial Dysfunction, Toxins 18(2):113
- 2025 – **Shihana F**, et al. Nephrotoxicity biomarkers for Propanil self-poisoning. Clin Toxicol (Phila).
- 2024 – Perananthan V, **Shihana F**, et al. Intestinal injury in paracetamol overdose (ATOM-8). Gastro & Hepatology.
- 2023 – **Shihana F**, et al. MicroRNAs signature panel identifies heavy drinkers with alcohol-associated cirrhosis. Biology, 12(10), 1314.
- 2023 – **Shihana F**, et al. MicroRNAs signature panel identifies heavy drinkers with cirrhosis. Preprint. DOI: 10.20944/preprints202309.1004.v1
- 2023 – **Shihana F**, et al. Lipid genes in fatty liver disease using zebrafish models. Liver Int. DOI: 10.1111/liv.15716
- 2023 – **Shihana F**, et al. Lipid gene role in fatty liver via zebrafish models. BioRxiv.
- 2022 – Dhanarisi J, **Shihana F**, et al. Alcohol co-ingestion and outcomes in pesticide poisoning. Alcohol & Alcoholism, 58(4):254-61.
- 2021 – **Shihana F**, et al. Urinary microRNAs as biomarkers for toxic acute kidney injury. Sci Rep, 11(1):9165.
- 2021 – **Shihana F**, et al. MicroRNAs in toxic acute kidney injury: Scoping review. Pharmacol Res Perspect, 9(2):e00695.
- 2020 – **Shihana F**, et al. Urinary vs. serum microRNAs in oxalic acid poisoning. Toxicol Lett, 334:21-6.
- 2020 – Wijerathna TM, **Shihana F**, et al. Oxidative stress in herbicide poisoning. Environ Toxicol Pharmacol, 80:103510.
- 2020 – Albuquerque P, **Shihana F**. Toxic acute kidney injury. Tropical Nephrology, 5:47-65.
- 2020 – **Shihana F**, et al. Circulating microRNA biomarkers in oxalic acid-induced AKI. Arch Toxicol, 94(5):1725-37.
- 2019 – Dhanarisi J, **Shihana F**, et al. Rocuronium to reduce ventilation duration in OP poisoning. Clin Toxicol (Phila), 58(4):254-61.
- 2019 – Albuquerque P, **Shihana F**, et al. AKI by Bothrops venom. Toxins (Basel), 11(3).
- 2018 – Jayasekara J, **Shihana F**, et al. Cystatin C vs. Creatinine in proteinuria. Malays J Med Sci, 25(6):67-75.
- 2018 – **Shihana F**, et al. Evaluation of "ChE check mobile" in AChE measurement. Clin Toxicol (Phila), 57(6):411-4.
- 2018 – Wijerathna TM, **Shihana F**, et al. Biomarkers for AKI from oxalic acid. Toxicol Lett, 299:182-90.
- 2017 – Wijerathna TM, **Shihana F**, et al. Conflicting evidence of AKI using creatinine vs. cystatin C. Clin Toxicol (Phila), 55(9):970-6.
- 2017 – Ratnayake I, **Shihana F**, et al. WBCT test for Russell's viper envenoming. Thromb Haemost, 117(3):500-7.
- 2016 – Hulse E, **Shihana F**, Buckley NA. Radical 7 co-oximeter inaccuracies. Clin Toxicol (Phila), 55(2):158.
- 2016 – **Shihana F**, Dawson AH, Buckley NA. Bedside test for methemoglobinemia. Bull WHO, 94(8):622-5.
- 2016 – **Shihana F**, Dawson AH, et al. Test improved antidote use in propanil poisoning. Clin Toxicol (Phila), 54(7):576-80.
- 2016 – Mohamed F, **Shihana F**, et al. Early biomarkers of GPSH nephrotoxicity. Toxicol Lett, 258:1-10.
- 2015 – Mohamed F, **Shihana F**, et al. Biomarkers for AKI and mortality in paraquat poisoning. Toxicol Lett, 237(2):140-50.
- 2015 – Mohamed F, **Shihana F**, et al. Rapid creatinine rise in paraquat poisoning. PLoS One, 10(3):e0122357.
- 2011 – **Shihana F**, et al. Stabilizing blood for methemoglobin testing. J Clin Lab Anal, 25(5):366-8.
- 2010 – **Shihana F**, et al. Colorimetric method for paracetamol. Clin Toxicol (Phila), 48(1):42-6.
- 2010 – **Shihana F**, et al. Bedside test for methemoglobin. Ann Emerg Med, 55(2):184-9.

PROFESSIONAL MEMBERSHIPS/ COMMITTEES

- Scientific Committee Member, Asia Pacific Association Medical Toxicology- 2022-2025
- Deputy Chair of Inclusion Gender Equity Program (IGEP) -Jan 2022- Dec 2022
- Committee member – Inclusion Gender Equity Program (IGEP) -Mar 2021-Jan 2022
- Member of Franklin Women (2020 – Present)
- Lifetime Member of Asia Pacific Association of Medical Toxicology (2008 – Present)

CONFERENCES PRESENTATIONS

- 2022 – ISBRA-ESBRA, Poland: Zebrafish models in fatty liver (Poster)
- 2022 – ISBRA-ESBRA, Poland: MicroRNAs in alcohol-related cirrhosis (Poster)
- 2019 – APAMT, Malaysia: Urinary microRNAs in AKI (Oral)
- 2019 – WCN, Melbourne: Circulating microRNAs in poisoning-induced AKI (Poster)
- 2018 – APAMT, Indonesia: Serum microRNAs in chemical-induced AKI (Oral)
- 2018 – ASN, California: Bothrops venom-induced AKI (Poster)
- 2018 – ANZSN, Sydney: Urinary/serum microRNAs in AKI (Mini Oral)
- 2018 – Hunter Cell Biology, Sydney: Novel AKI biomarkers (Poster)
- 2016 – APAMT, Singapore: Antioxidant therapy in paraquat injury (Oral)
- 2015 – APAMT, Perth: "ChE check mobile" accuracy in AChE measurement (Oral)
- 2015 – APAMT, Perth: NGAL in AKI from Russell's viper bites (Oral)
- 2015 – APAMT, Perth: Acute poisoning patterns in Sri Lanka (Oral)
- 2015 – WCN, South Africa: GSH-induced AKI biomarkers (Poster)
- 2014 – APAMT, China: Cystatin C vs. creatinine in CKD (Oral)
- 2013 – APAMT, Dubai: Early creatinine rise in paraquat poisoning (Oral)
- 2013 – WCN, Hong Kong: AKI biomarkers in GSH poisoning (Poster)
- 2012 – Peradeniya Symposium: Cyanobacterial toxin impact on mouse kidneys (Oral)